Inside the Atom: Protons, Neutrons and Electrons

Answer key

A. The particles

1. Charge, mass and location.

> Proton: charge +1, mass about 1 u, in the nucleus.
>
> Neutron: charge 0, mass about 1 u, in the nucleus.
>
> Electron: charge −1, mass about 1/1836 of a proton, around the nucleus.

2. Atomic number and mass number.

> Atomic number = number of **protons** (identifies the element). Mass number = number of **protons** + **neutrons**.

B. Counting particles

3. Protons, neutrons, electrons.

	Protons	Neutrons	Electrons
(a) F	9	10	9
(b) K^+	19	20	18
(c) S^{2-}	16	16	18
(d) Al^{3+}	13	14	10
(e) Zn	30	34	30

4. Write the symbols.

> (a) 26 protons = iron; A = 26 + 30 = 56; 24 electrons gives 2+: $^{56}_{26}Fe^{2+}$
>
> (b) 8 protons = oxygen; A = 8 + 10 = 18; 10 electrons gives 2−: $^{18}_{8}O^{2-}$

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C. Isotopes

5. Neutrons in hydrogen isotopes.

> Hydrogen-1: **0**; hydrogen-2: **1**; hydrogen-3: **2** (each has 1 proton).

6. Why isotopes behave the same chemically.

> They have the same number of protons and therefore, as neutral atoms, the same number of **electrons**. Chemical behaviour depends on the electrons, not the neutrons.

D. Challenge

7. Why 35.45?

> 35.45 is the **weighted average** mass of chlorine's naturally occurring isotopes (mainly chlorine-35 and chlorine-37). Mass numbers are whole-number counts for individual isotopes.

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